

# User Manual (How to use the program)

- This application is still under development. At this stage, you may need to launch or run the application in developer mode to use it.

## Prerequisites

### How to Locate Your Alpaca API Keys:

#### 1. Log in to Your Alpaca Account

Navigate to the Alpaca website and sign in with your registered username and password.

#### 2. Access the Dashboard

After logging in, you will be taken to the main dashboard. If not, select **Dashboard** from the menu.

#### 3. Open the API Keys Section

In the left-hand navigation panel, select **API Keys** or **Your API Keys**, depending on your account view.

#### 4. Select the Appropriate Environment

Choose between:

- **Paper Trading** (for testing)
  - **Live Trading** (for real transactions)
- Each environment has its own set of keys.

#### 5. Reveal the Keys

Click **Generate** (if no keys exist) or **View/Reveal** (if keys are already created).

You will see two values:

- **API Key ID**
- **Secret Key**

#### 6. Copy and Store the Keys Securely

Copy both keys and store them in the *settings.toml* file under *[keys]* section with quote marks around the keys ("\*key\*").

*Do not share these keys publicly.*

#### 7. Regenerate if Needed

If you suspect your keys have been exposed, you can select **Regenerate** to create new ones.

### How to Install the Required Python Packages:

Follow these short steps to set up your environment:

#### 1. Install Pythonpython --version

- Make sure Python 3.11 or newer (but < 4.0) is installed:

#### 2. Install uvcurl -LsSf <https://astral.sh/uv/install.sh> | shuv --version

- If you do not already have uv installed:
- Verify installation
- 3. Navigate to backendcd backend
- 4. Create a Virtual Environmentuv venvsource .venv/bin/activate
  - Activate the environment
- 5. Install dependenciesuv syncuv lock --upgradeuv sync
  - This will install the locked dependencies from uv.lock:
  - To update dependencies to newer compatible versions:
- 6. Run backend APIuv run main.py

## How to open app

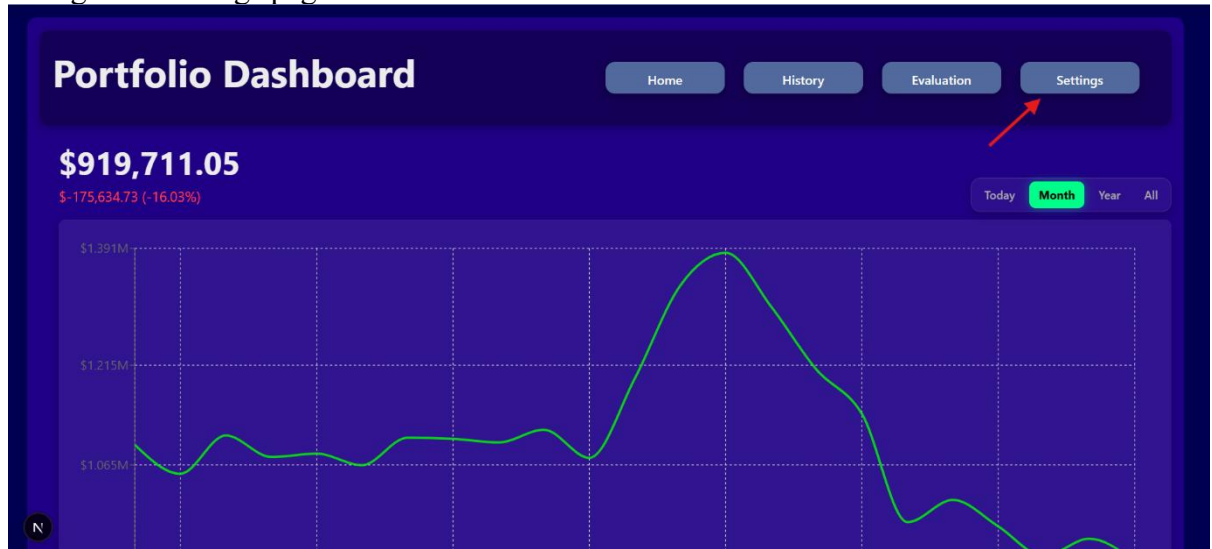
Follow the short steps to open your app in another terminal while the first terminal runs (DO NOT CLOSE THE FIRST TERMINAL):

- 1. Navigate to frontendcd ../frontendcd frontend
  - If you are located in backend run:
  - If you are located in the root file run:
- 2. Install dependenciesnpm install
- 3. Run application with:npm run dev
- 4. Open your favorite web browser and enter the following URL in the address bar to open the application:
  - <http://localhost:3000/>

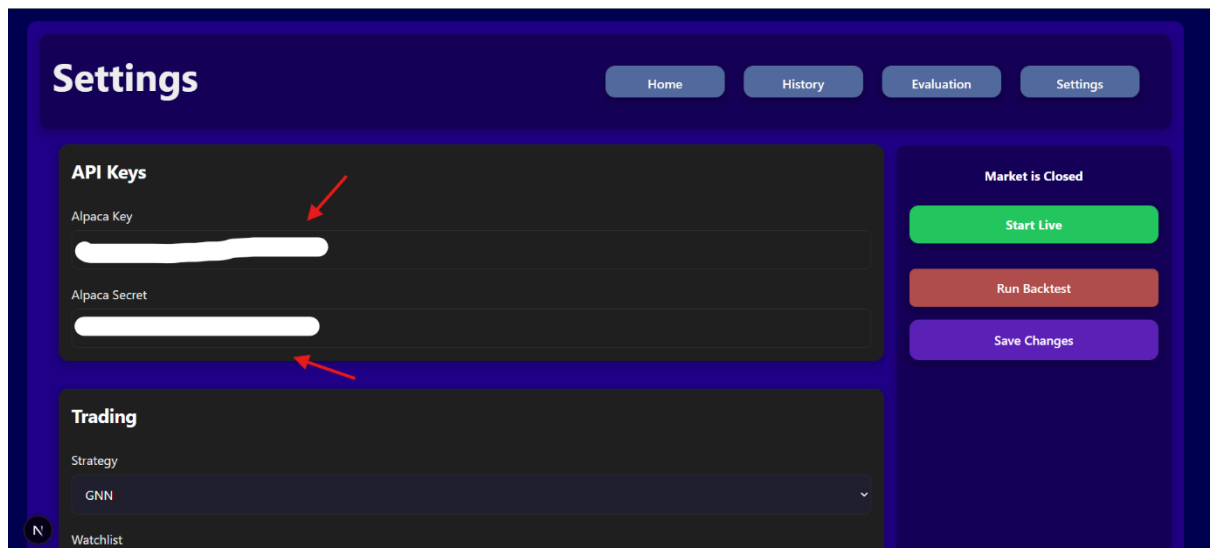
## User Guide:

The steps below explain how to use the app:

1. Navigate to settings page

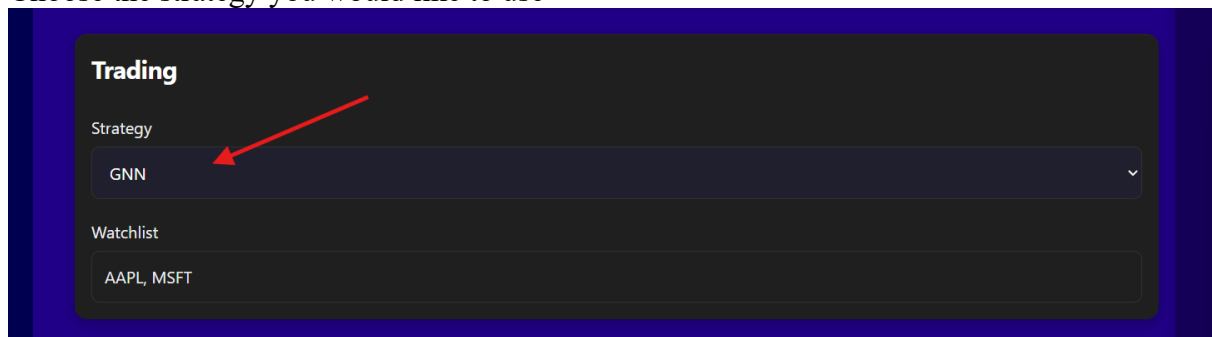


2. Paste your Alpaca key and secret-key



The screenshot shows the 'Settings' page with a dark blue theme. At the top, there are navigation buttons: Home, History, Evaluation, and Settings. The 'Settings' button is highlighted. Below the navigation bar, there are two main sections: 'API Keys' and 'Trading'. In the 'API Keys' section, there are two input fields: 'Alpaca Key' and 'Alpaca Secret'. Both fields are filled with white text and have a red arrow pointing to them. In the 'Trading' section, there is a 'Strategy' dropdown menu with 'GNN' selected. Below the 'Strategy' dropdown is a 'Watchlist' input field containing the text 'AAPL, MSFT'. To the right of these sections, there is a 'Market is Closed' status and three buttons: 'Start Live' (green), 'Run Backtest' (red), and 'Save Changes' (purple).

3. Choose the strategy you would like to use

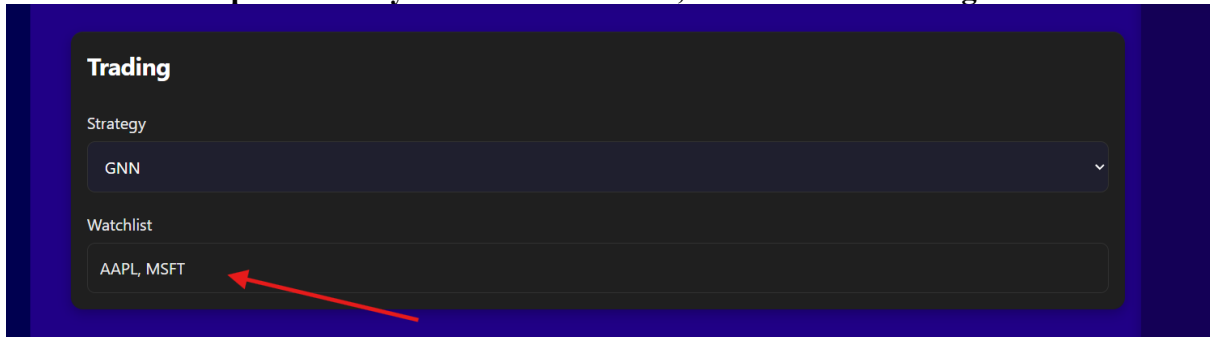


The screenshot shows the 'Trading' section of the settings page with a dark blue theme. It features a 'Strategy' dropdown menu with 'GNN' selected. A red arrow points to the 'GNN' option. Below the 'Strategy' dropdown is a 'Watchlist' input field containing the text 'AAPL, MSFT'.

4. Enter the stock symbols you want the AI to analyze

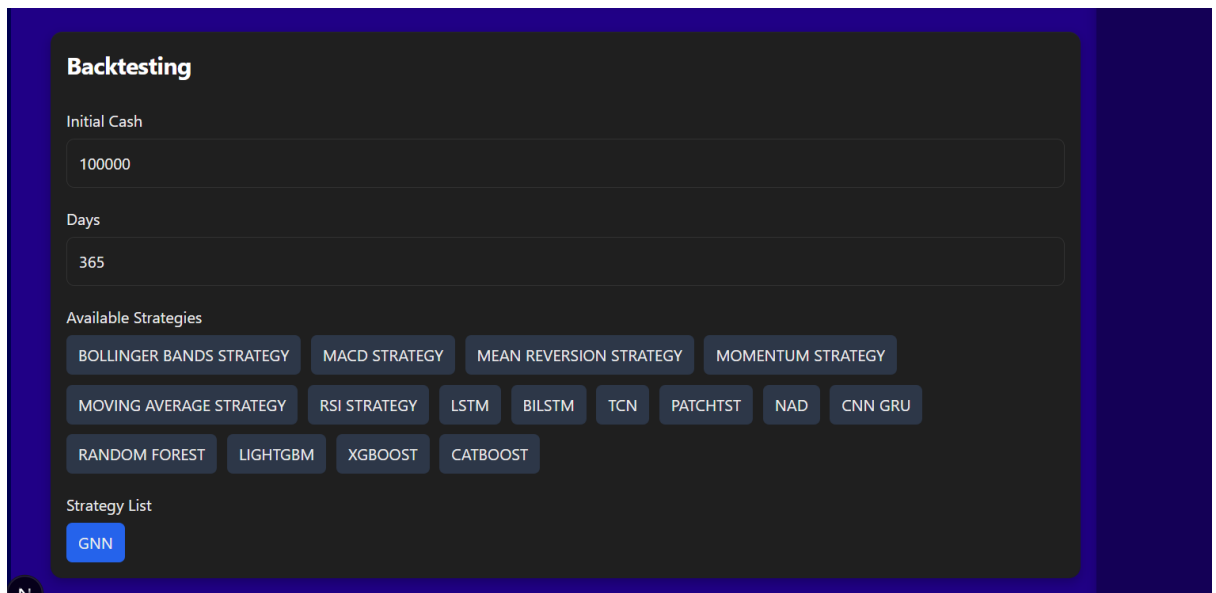
- If this list is empty, the AI will only make predictions for stocks in which you already hold a position.

- Remember to separate the symbols with commas, as shown in the image below:



5. In the Backtesting card, you can configure the following settings:

1. Initial Cash: the amount of cash available at the start of the backtest.
2. Days: the number of days to include in the backtest, counting backward from today.
3. Strategies: click a strategy to add it to the backtest. Click it again to remove it from the list.



6. These settings control the AI configuration. Adjust them as needed, although the default values are suitable for most users

- *All values must be entered as integers.*

**ML Settings**

ML Training Lookback

750

SMA 1

5

SMA 2

20

SMA 3

50

RSI Window

14

MACD Fast

12

MACD Slow

26

MACD Signal

9

Time Steps

50

ZScore Window

100

- Remember to save changes!

**Settings**

Home History Evaluation Settings

**API Keys**

Alpaca Key

PKM5ZULBDBE3ECEY2LRQLHMNUW

Alpaca Secret

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**Trading**

Strategy

GNN

Watchlist

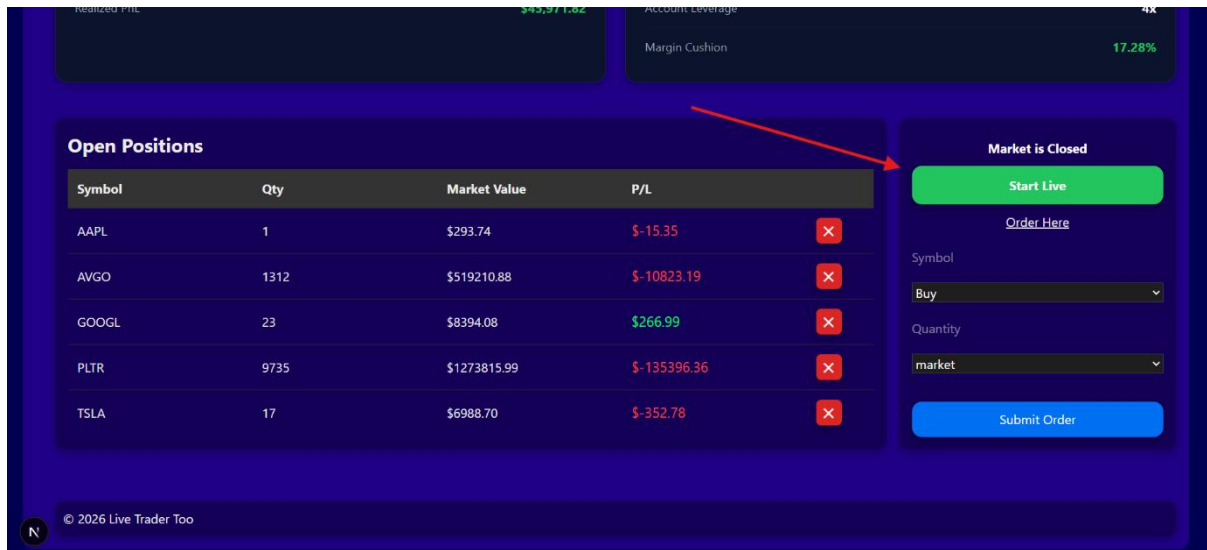
Market is Closed

Start Live

Run Backtest

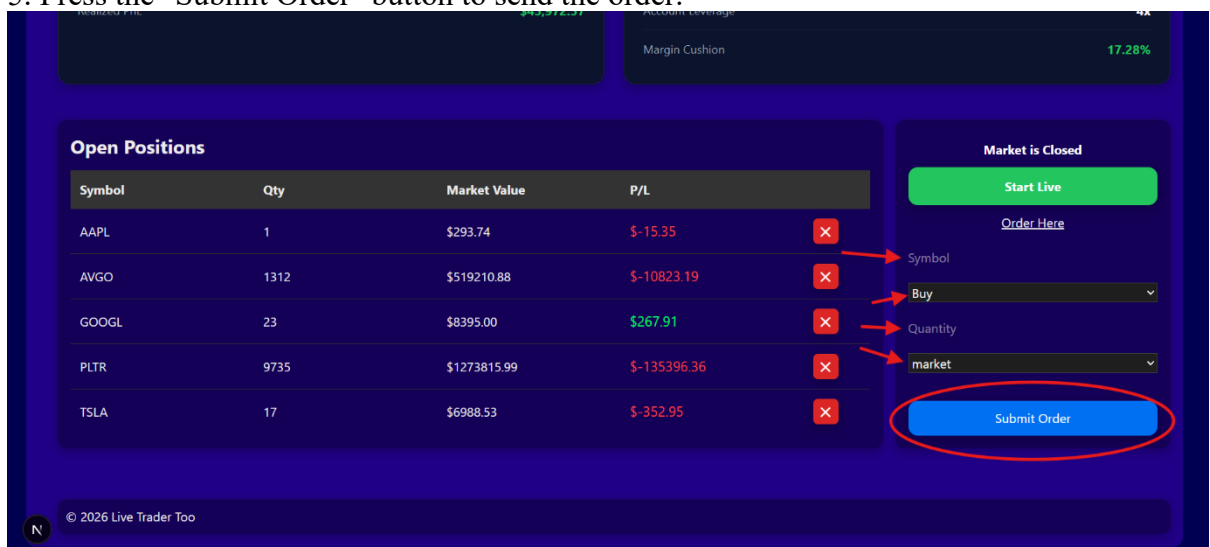
Save Changes

7. On all pages, you will find a green button that starts live automated trading when clicked. Click the button again to stop live trading



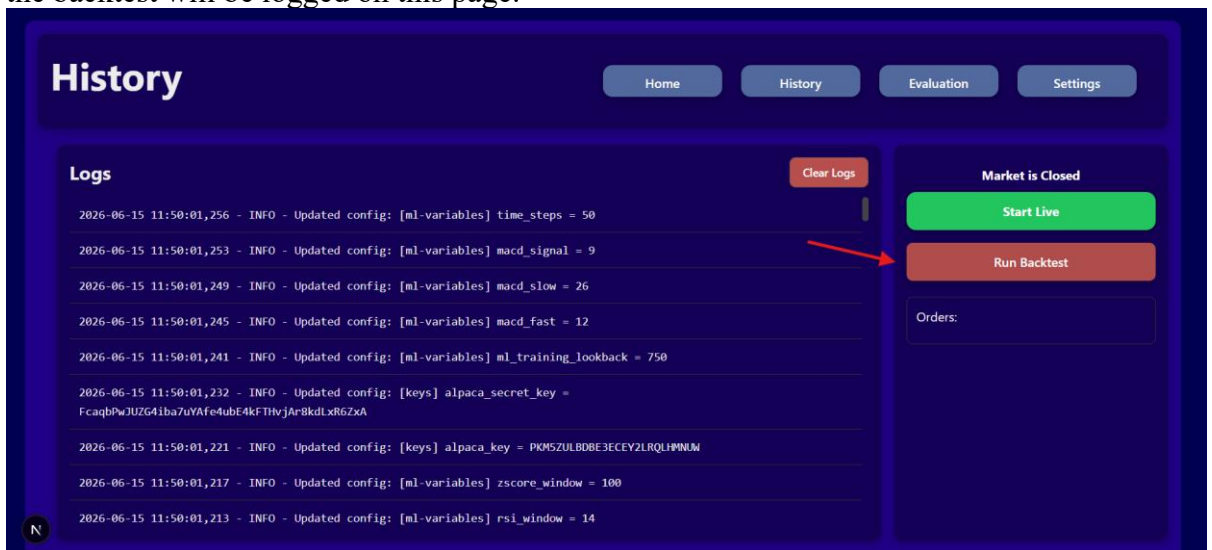
8. Below the equity chart located on the Home page there exists a ordering panel

1. Enter the symbol of the stock you want to place an order in.
2. Choose from buying or selling order.
3. Enter the amount of quantities you want to order.
4. Choose the order type.
5. Press the "Submit Order" button to send the order.



9. The History page contains logs related to AI decisions, orders, and any errors that occur

- You will also find the Run Backtest button. The results and information generated by the backtest will be logged on this page:



10. The Evaluation page contains the result of the backtesting and model-evaluations.

